

Database Lab Questions for LAB TEST.

Date: 11/01/2023

Q. NO.	SCHEMA and Problem Statements
1	Airline Database: AIRCRAFT (<u>Aircraft ID</u>, Aircraft_name, Cruising_range) CERTIFIED (<u>Emp ID</u>, Aircraft ID) EMPLOYEE (<u>Emp ID</u>, Ename, Salary) i. Find the employee ID's of employee who make the highest salary ii. Find the employees who are not certified for operating any aircraft
2	Employee database: EMPLOYEE (Fname, Lname, <u>SSN</u>, Addr, Sex, Salary, SuperSSN, Dno) DEPARTMENT (Dname, Dnumber, MgrSSN, MgrStartDate) i. For each department, retrieve the department name and the average salary of all employees working in that department ii. Display the details of all departments having 'tech' as their substring
3	Car Ownership database: PERSON (<u>driver - id #</u>, name, address) CAR (<u>Regno</u>, model, year) OWNS (<u>driver-id #</u>, <u>Regno</u>) i. Display the number of cars owned by each driver. 2. Find the address of driver having id has 115
4	Cricket database: PLAYER (Pid, Lname, Fname, Country, Yborn, Bplace) MATCH (MatchId, Team1, Team2, Ground, Date, Winner) i. Display the sorted list of ground names where India has played as team1 2. Find name of the team who won match on 10 Dec 2023
5	Library Database: Student(<u>Stud_no</u> : integer, Stud_name: string) Membership(Mem_no: integer, Stud_no: integer) i. List all the student names with their membership numbers ii. create a view to print the total no. of students and membership available.
6	Project database: STUDENT (Rollno: integer, Name: String, Sem: integer, Guide_No: integer) GUIDE (Guide_name: String, Guide_No: integer, Guide_research_domain: String,) PROJECT (Project_No: Integer, Project_title: String, Project_Area: String, Guide_No: integer) i. Find the list of project no, project name & name of guide, in domain of DataBase.
7	Employee Database: Employee (F_name: string, L_name: string, SSN: integer, Salary: integer, Super_Emp_id: integer, D_no: integer) Department (D_name: string, D_no: integer, D_Mgr_id: integer, Mgr_start_date: date) i. List all the manager's details. 2. List all the employee's other than managers.

8	Sailors(sid: integer, sname: string, rating: integer, age: real); Boats(bid: integer, bname: string, color: string); Reserves(sid: integer, bid: integer, day: date) I. Find the names of sailors who have reserved a red boat, and list in the order of age .ii.) Find the names of sailors who have reserved boat 103
9	TRAIN Reservation System Train(No. Name,From, To, SeatsAvailable) Passenger(PID, TrainNO, From, To, no.of seats booked) I. list out Train details which is empty to full order(seats) li. list out train details which are booked by passengers lii. list out train details which are not booked by any passengers.
10	Bank Database: Bank(ID, Name,Location) Customer(CID, BankID, Name,Balance) I. list out the bank who has atleast one customer. li. list out the customer who is having account in all banks.
11	Consider the following relational schemas: EMPLOYEE (EMPLOYEE_NAME, STREET, CITY) WORKS (EMPLOYEE_NAME, COMPANYNAME, SALARY) COMPANY (COMPANY_NAME, CITY) I. Display all employee and companies present in City ='Bengaluru' li. Display all employee working for a company, arrange it company wise.
12	Doctor(DName,Reg_no) Patient(Pname, Disease) Assigned_To (Pname,Dname) I. Get the names of doctors who are treating patients with 'COVID' li. Find how many patients are treated by Doctor "XYZ". lii. Display the doctor details , arrange it by Reg_no.
13	SALESPERSON (SSN, Name, start_year, Dept_no) TRIP (SSN, From_city, To_city, Trip_ID) EXPENSE(TripID, Account#, Amount) I. Display the expenses, From_city and salesperson information of TripID='777'
14	Given the following relations TRAIN (NAME, START, DEST) TICKET (PNRNO., START, DEST, FARE) PASSENGER (NAME, ADDRESS, PNRNO.) I. find all trains between Bengaluru to NewDelhi li. Find the train name booked by Passenger ='ABC'.
15	MOVIES(Title, YoP,Length). STARS(Name,ID,) ACT(MovieTitle,StarName) I. Print all movies released in chronological order. II. Print all movies acted by 'KISHORE'